



Hands-on No. : 2

Topic : Exception Handling

Date : 15-12-25

1. Create a class *MyCalculator* which consists of a single method `long power(int, int)`. This method takes two integers, `n` and `p`, pass parameters and finds n^p . If either `n` or `p` is negative, then the method must throw an exception which says " `n` or `p` should not negative". Also, if both `n` and `p` are zero, then the method must throw an exception which says " `n` and `p` should not be Zero ".

Sample Input

3 5

2 4

0 0

-1 -2

-1 3

Sample Output

243

16

java.lang.Exception: `n` and `p` should not be zero.

java.lang.Exception: `n` or `p` should not be negative.

java.lang.Exception: `n` or `p` should not be negative.

2. Accept Username and password values from the User . Throw user defined exceptions – "InvalidUsernameException" or "InvalidPasswordException" if the username and password are invalid .

Please check with Stored username and password for validity.

If valid, print a message "Welcome 'username'", else print a message Invalid username or password.

A username is considered valid if all the following constraints are satisfied:

- a) The username consists of 6 to 30 characters inclusive. If the username consists of less than 6 or greater than 30 characters, then it is an invalid username.

It is going to be hard but, hard does not mean impossible.



- b) The username can only contain alphanumeric characters and underscores (_). Alphanumeric characters describe the character set consisting of lowercase characters [a – z], uppercase characters [A – Z], and digits [0 – 9].
- c) The first character of the username must be an alphabetic character, i.e., either lowercase character [a – z] or uppercase character [A – Z].

A password is considered valid if all the following constraints are satisfied:

- a) It contains at least one lowercase English character.
- b) It contains at least one uppercase English character.
- c) It contains at least one special character. The special characters are:
!@#\$%^&*()-+
- d) Its length is at least 8.
- e) It contains at least one digit.

3. Define an employee class with properties Employee code, name, date of birth and date of appointment. The Employee code must be a positive integer number.
 - Write a Java program to read the above details and validate the employee code. If the employee code is not in the format specified, then raise an exception called **InvalidEmpNumberException**.
 - Verify if the date of birth is before the data of appointment. If it is not so then raise an exception called **InvalidDateOfJoinException**. If it is correct, then create the Employee object and display the details of employees and the number of years of experience.
4. Create a java class for handling an Exception called 'PayOutOfBoundsException' and throw the exception when the transaction amount exceeds the limit or the amount is insufficient. (Maximum transaction limit is 30000). Create a class called 'AccountManagement' with two methods named 'checkForDebit' and 'withdrawAmount' that uses PayOutOfBoundsException. (Keep Current balance as 80000).
5. Create a class called Invoice that a Grocery store might use to represent an invoice for an item sold at the store. An Invoice should include four pieces of information as instance variables—a part number (type integer), a part description (type

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String), quantity of the item being purchased (type integer) and a price per item (double). Provide method constructor with four arguments. Write a test application to create an instance and validate the input obtained using Scanner object. Ensure that part number is value greater than 0, part description is not null string, quantity of the item and price per item is value greater than 0.

Note : The InputMismatchException is thrown when attempting to retrieve a value using the Scanner class that doesn't match the expected pattern or type.

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